

		Q since the work done on the gas W is negative.
16	A	Maximum potential energy = $\frac{1}{2}m\omega^2x^2 = E$. The new maximum potential energy is $\frac{3}{4}E$ after the maximum potential energy of the object is reduced by $\frac{1}{4}E$. $\frac{\frac{1}{2}m\omega^2x^2}{\frac{1}{2}m\omega^2x_0^2} = \frac{E}{\frac{3}{4}E}$ where x_0 is the new amplitude. $\Rightarrow x_0 = 0.866 x$ Hence change in amplitude = $x - x_0 = x - 0.866 x = 0.134 x$
17	C	